

Optical cum radar imaging satellite for better Earth-monitoring

A new OptoSAR satellite built by the space-tech startup GalaxEye, designed to combine optical and radar imaging for continuous planetary observation, is being prepared for launch, aiming to deliver clearer, more reliable Earth-



GalaxEye to launch world's first OptoSAR satellite with Nvidia Jetson Orin (Credit: www.livemint.com)

monitoring data. The satellite integrates electro-optical cameras with synthetic aperture radar on a single platform, enabling high-resolution imaging even under cloud cover, at night, or in adverse weather. The system is being developed in Bengaluru and will also use onboard AI computing to process observation data directly in orbit. The technology is expected to support applications such as disaster monitoring, agricultural analysis, environmental tracking, and infrastructure surveillance.

Innovative blood-clotting detection to help design safer medical devices

Researchers at the Indian Institute of Technology Madras have developed an optics-based technique to precisely detect blood clot formation by monitoring subtle changes in reflected light, offering a new way to evaluate how medical materials interact with blood. The approach tracks variations in surface reflectivity and turbidity as blood contacts implant materials, enabling measurement of clotting time with millisecond-level precision. The method helps assess haemocompatibility, critical for devices such as stents, heart valves, and catheters. By enabling faster, more accurate testing of implant materials, the innovation could support safer medical device design and reduce clot-related complications in clinical settings.

Chip-fabrication technique that strengthens cryptographic hardware

Engineers at the Massachusetts Institute of Technology (MIT) have developed a chip-fabrication technique that strengthens cryptographic hardware by embedding matching physical fingerprints in paired microchips, enabling them to authenticate each other without storing secret keys externally. The approach creates a shared physical unclonable function during manufacturing, exploiting microscopic variations in CMOS processes to generate a unique,

hard-to-replicate hardware identity. The twin-chip system demonstrated over 98% reliability in matching these fingerprints, allowing secure, server-free authentication. The technology was developed by researchers for energy-constrained devices such as IoT nodes and medical sensors, requiring lightweight yet robust hardware security.

'Negative light' communication nearly impossible to detect

Scientists at UNSW Sydney, in collaboration with Monash University, have demonstrated a communication method using 'negative light' that hides data transmission within



A thermoradiative diode which is able to transmit hidden data (Credit: UNSW)

natural heat radiation, making the signal nearly impossible to detect. The technique exploits negative luminescence, in which infrared emitters appear darker rather than brighter, allowing data to blend into ambient thermal radiation while remaining readable by specialised receivers. Early lab tests achieved speeds of

about 100kB/s, with potential for gigabit-per-second performance as devices improve. The system offered a potential breakthrough in secure communications for sectors such as defence, finance, and critical infrastructure.

Organic cathode pushes battery limits

A newly engineered organic cathode could push lithium-ion batteries beyond current performance limits by improving energy density, safety, and temperature resilience. The technol-



Chinese researchers' breakthrough in organic battery technology creates a safer, more durable alternative (Credit: <https://en.tju.edu.cn>)

ogy was developed by researchers at Tianjin University in collaboration with South China University of Technology. The design uses a conductive polymer cathode, poly-benzofurandione (PBFDO), enabling batteries to exceed 250Wh/kg, higher than typical lithium iron phosphate cells. The bat-

ttery also operates over an unusually wide temperature range, from -70°C to 80°C, while maintaining structural stability under bending and compression. Scientists are advancing organic electrode materials as safer, resource-efficient alternatives for flexible electronics and wearable power systems.